

# Shengwei ZHOU

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✉ s.arthur.zhou@gmail.com • 🌐 [sites.google.com/view/shengweizhou](https://sites.google.com/view/shengweizhou)

## Research interests

Fair Division, Algorithmic Game Theory, Mechanism Design, Online Algorithm, Approximation Algorithm.

## Education

- 2021 – 2025 **Ph.D. in Computer Science**, *University of Macau*, Macao S.A.R.  
Thesis: “Fairly Allocating Indivisible Chores to Symmetric and Asymmetric Agents”  
Supervisor: Prof. Xiaowei Wu
- 2019 – 2020 **M.Sc. in Urban Informatics**, *King’s College London*, London, UK  
Dissertation Supervisor: Dr. Angus Roberts
- 2014 – 2019 **B.Eng. in Urban-rural Planning**, *Wuhan University*, Wuhan, China  
Minor in Geography Information Science

## Research experience

- 2025 – Present **Research Fellow**, *Nanyang Technological University*, Singapore  
Supervisor: Prof. Xiaohui Bei
- 2021 **Research Assistant**, *University of Macau*, Macau S.A.R.  
Supervisor: Prof. Xiaowei Wu

## Honors and scholarships

- 2025 **Best Student Paper Award**, *Conference on Web and Internet Economics (WINE)*  
1 out of 225 submissions
- 2025 **Taihu Scholarship**, *University of Macau*  
The Guangdong-Hong Kong-Macao Greater Bay Area Taihu Scholarship
- 2021 – 2025 **UM Ph.D. Assistantship**, *University of Macau*
- 2019 **Outstanding Bachelor’s Thesis Award**, *Wuhan University*  
Top 5%, School of Urban Design

## Working papers ( $\alpha$ - $\beta$ ): Alphabetical order   ✉: Corresponding author

- [W5] **Subsidised Proportionality in Fair Allocation**  
( $\alpha$ - $\beta$ ) Xiaowei Wu, Cong Zhang, Shengwei Zhou✉.  
*Minor revision at Operations Research*
- [W4] **Degree-bounded Online Bipartite Matching Revisited: OCS vs. Ranking**  
( $\alpha$ - $\beta$ ) Yilong Feng, Haolong Li, Xiaowei Wu, Shengwei Zhou.  
*Submitted to Mathematics of Operations Research*

- [W3]    **Efficient EFX Allocations for Chores**  
 $(\alpha\text{-}\beta)$  Zehan Lin, Xiaowei Wu, Shengwei Zhou.  
*Under review at Artificial Intelligence*
- [W2]    **An FPTAS for 7/9-Approximation to Maximin Share Allocations**  
 $(\alpha\text{-}\beta)$  Xin Huang, Shengwei Zhou.  
*Working paper*
- [W1]    **When Maximum Nash Welfare Becomes Strongly Fair: Bi-valued Goods**  
 $(\alpha\text{-}\beta)$  Zehan Lin, Xiaowei Wu, Shengwei Zhou.  
*Working paper*

**Conference proceedings**     $(\alpha\text{-}\beta)$ : Alphabetical order

- [C17]    **Fair Division by Contribution: A Shapley Value Perspective**  
 $(\alpha\text{-}\beta)$  Xiaohui Bei, Pinyan Lu, Xiaowei Wu, Shengwei Zhou.  
*In Proceedings of the 27th ACM Conference on Economics and Computation (EC), 2026, Forthcoming.*
- [C16]    **Allocating Chores with Restricted Additive Costs: Achieving EFX, MMS, and Efficiency Simultaneously**  
 $(\alpha\text{-}\beta)$  Zehan Lin, Xiaowei Wu, Shengwei Zhou.  
*In Proceedings of the ACM on Web Conference (WWW), pages 146-156, Apr 2026.*  
doi:[10.1145/3774904.3792300](https://doi.org/10.1145/3774904.3792300)
- [C15]    **All-but-One MMS Allocation for Chores**  
 $(\alpha\text{-}\beta)$  Jiawei Qiu, Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
*In Proceedings of the ACM on Web Conference (WWW), pages 157-168, Apr 2026.*  
doi:[10.1145/3774904.3792305](https://doi.org/10.1145/3774904.3792305)
- [C14]    **Degree-bounded Online Bipartite Matching Revisited: OCS vs. Ranking**  
 $(\alpha\text{-}\beta)$  Yilong Feng, Haolong Li, Xiaowei Wu, Shengwei Zhou.  
*In Proceedings of the 21st Conference on Web and Internet Economics (WINE), pages 747–764, Dec 2025.* doi:[10.1007/978-3-032-18660-7\\_39](https://doi.org/10.1007/978-3-032-18660-7_39) ★ **Best Student Paper**
- [C13]    **Incentive Analysis of Collusion in Fair Division**  
 $(\alpha\text{-}\beta)$  Haoqiang Huang, Biaoshuai Tao, Mingwei Yang, Shengwei Zhou.  
*In Proceedings of the 21st Conference on Web and Internet Economics (WINE), pages 392–410, Dec 2025.* doi:[10.1007/978-3-032-18660-7\\_21](https://doi.org/10.1007/978-3-032-18660-7_21)
- [C12]    **When is Truthfully Allocating Chores no Harder than Goods?**  
 $(\alpha\text{-}\beta)$  Bo Li, Biaoshuai Tao, Fangxiao Wang, Xiaowei Wu, Mingwei Yang, Shengwei Zhou.  
*In Proceedings of the 18th International Symposium on Algorithmic Game Theory (SAGT), pages 247-264, Sep 2025.* doi:[10.1007/978-3-032-03639-1\\_14](https://doi.org/10.1007/978-3-032-03639-1_14).

- [C11] **Approximately EFX and PO Allocations for Bivalued Chores**  
 $(\alpha\text{-}\beta)$  Zehan Lin, Xiaowei Wu, Shengwei Zhou.  
*In Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJ-CAI)*, pages 3952-3960, Aug 2025. doi:[10.24963/ijcai.2025/440](https://doi.org/10.24963/ijcai.2025/440).
- [C10] **A Little Subsidy Ensures MMS Allocation for Three Agents**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Quan Xue, Shengwei Zhou.  
*In Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJ-CAI)*, pages 4073-4081, Aug 2025. doi:[10.24963/ijcai.2025/454](https://doi.org/10.24963/ijcai.2025/454).
- [C9] **Revisiting Proportional Allocation with Subsidy: Simplification and Improvements**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Quan Xue, Shengwei Zhou.  
*In Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJ-CAI)*, pages 4082-4090, Aug 2025. doi:[10.24963/ijcai.2025/455](https://doi.org/10.24963/ijcai.2025/455).
- [C8] **Pure Nash Equilibrium of Weight Picking Sequence Protocol is WEF1 for Two Agents**  
 $(\alpha\text{-}\beta)$  Rufan Bai, Huahua Miao, Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
*In Proceedings of the 19th International Joint Conference, IJTCS-FAW*, pages 238-251, Jun 2025. doi:[10.1007/978-981-96-8312-3\\_18](https://doi.org/10.1007/978-981-96-8312-3_18).
- [C7] **Tree Splitting Based Rounding Scheme for Weighted Proportional Allocations with Subsidy**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Shengwei Zhou.  
*In Proceedings of the 20th Conference on Web and Internet Economics (WINE)*, pages 295-313, Dec 2024. doi:[10.1007/978-3-032-08560-3\\_17](https://doi.org/10.1007/978-3-032-08560-3_17).
- [C6] **On the Existence of EFX (and Pareto-Optimal) Allocations for Binary Chores**  
 $(\alpha\text{-}\beta)$  Biaoshuai Tao, Xiaowei Wu, Ziqi Yu, Shengwei Zhou.  
*In Proceedings of the 18th International Joint Conference, IJTCS-FAW*, pages 33-52, Aug 2024. doi:[10.1007/978-981-97-7752-5\\_3](https://doi.org/10.1007/978-981-97-7752-5_3).
- [C5] **One Quarter Each (on Average) Ensures Proportionality**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
*In Proceedings of the 20th Conference on Web and Internet Economics (WINE)*, pages 582-599, Dec 2023. doi:[10.1007/978-3-031-48974-7\\_33](https://doi.org/10.1007/978-3-031-48974-7_33).
- [C4] **Improved Competitive Ratio for Edge-Weighted Online Stochastic Matching**  
Guoliang Qiu, Yilong Feng, Shengwei Zhou, Xiaowei Wu.  
*In Proceedings of the 20th Conference on Web and Internet Economics (WINE)*, pages 527-544, Dec 2023. doi:[10.1007/978-3-031-48974-7\\_30](https://doi.org/10.1007/978-3-031-48974-7_30).
- [C3] **Multi-agent Online Scheduling: MMS Allocations for Indivisible Items**  
Shengwei Zhou, Rufan Bai, Xiaowei Wu.  
*In Proceedings of the 40th International Conference on Machine Learning (ICML)*, pages 42506-42516, Jul 2023. doi:[10.5555/3618408.3620197](https://doi.org/10.5555/3618408.3620197).

- [C2] **Weighted EF1 Allocations for Indivisible Chores**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
 In *Proceedings of the 24th ACM Conference on Economics and Computation (EC)*, pages 1155, Jul 2023. doi:[10.1145/3580507.3597763](https://doi.org/10.1145/3580507.3597763).
- [C1] **Approximately EFX Allocations for Indivisible Chores**  
 Shengwei Zhou, Xiaowei Wu.  
 In *Proceedings of the 31st International Joint Conference on Artificial Intelligence (IJ-CAI)*, pages 783-789, Aug 2022. doi:[10.24963/ijcai.2022/110](https://doi.org/10.24963/ijcai.2022/110).

### Journal publications $(\alpha\text{-}\beta)$ : Alphabetical order

- [J3] **Weighted EF1 Allocations for Indivisible Chores**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
*Artificial Intelligence (AIJ)*, 347:104386, Oct 2025. doi:[10.1016/j.artint.2025.104386](https://doi.org/10.1016/j.artint.2025.104386).  
 Preliminary version in EC 23 (C2).
- [J2] **On the Existence of EFX (and Pareto-Optimal) Allocations for Binary Chores**  
 $(\alpha\text{-}\beta)$  Biaoshuai Tao, Xiaowei Wu, Ziqi Yu, Shengwei Zhou<sup>✉</sup>.  
*Theoretical Computer Science (TCS)*, 1042:115248, Jul 2025.  
 doi:[10.1016/j.tcs.2025.115248](https://doi.org/10.1016/j.tcs.2025.115248). Preliminary version in IJTCS-FAW 24 (C6).
- [J1] **Approximately EFX Allocations for Indivisible Chores**  
 Shengwei Zhou, Xiaowei Wu.  
*Artificial Intelligence (AIJ)*, 326:104037, Jan 2024. doi:[10.1016/j.artint.2023.104037](https://doi.org/10.1016/j.artint.2023.104037).  
 Preliminary version in IJCAI 22 (C1).

### Teaching experience

- Fall 2022 **Teaching assistant**, *University of Macau*  
 – 2024 CISC7007 Design and Analysis of Algorithm (Postgraduate)
- Spring 2022 **Teaching assistant**, *University of Macau*  
 – 2025 CISC2006 Algorithm Design and Analysis (Undergraduate)
- Fall 2021 **Teaching assistant**, *University of Macau*  
 CISC3027 Special Topics in Computer and Information Science (Undergraduate)

### Talks and tutorials

- Jul 2026 Fair Division by Contribution: A Shapley Value Perspective  
*ACM EC, Rome*
- Aug 2025 Revisiting Proportional Allocation with Subsidy: Simplification and Improvements  
*IJCAI, Guangzhou*
- Apr 2025 Revisiting Proportional Allocation with Subsidy: Simplification and Improvements  
*IOTSC Postgraduate Forum, University of Macau*
- Oct 2024 Fair Allocations of Chores with Subsidy  
*Invited Talk, Kyushu University*

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| Aug 2024 | On the Existence of EFX (and Pareto-Optimal) Allocations for Binary Chores<br><i>IJTCS-FAW, Hong Kong</i>      |
| Mar 2024 | PROP Allocation for Indivisible Chores with Subsidy<br><i>Invited Talk, Shanghai Jiao Tong University</i>      |
| Dec 2023 | Weighted PROP Allocations with Subsidy<br><i>Workshop on Theoretical Computer Science, University of Macau</i> |
| Dec 2023 | Recent Development of Weighted Fair Allocation<br><i>Invited Talk, Wuhan University</i>                        |
| Aug 2023 | Weighted EF1 Allocations for Indivisible Chores<br><i>Young PhD Forum, IJTCS-FAW, Macao</i>                    |
| Jul 2023 | Weighted EF1 Allocations for Indivisible Chores<br><i>ACM EC, London</i>                                       |
| Nov 2022 | Approximately EFX Allocations for Indivisible Chores<br><i>IJCAI-China, Shenzhen</i>                           |

### Service and outreach

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|---------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>PC Member</b>                | AAMAS (2026), EC (2026), IJCAI (2026), WINE (2026), WINE (2025)                                             |
| <b>Conference (Sub)reviewer</b> | NeurIPS (2026), SODA (2026), WWW (2025), ISAAC (2024), SODA (2023)                                          |
| <b>Journal Reviewer</b>         | Journal of Autonomous Agents and Multi-Agent Systems (JAAMAS), SIAM Journal on Discrete Mathematics (SIDMA) |
| <b>Organizing Committee</b>     | IJTCS-FAW (2023)                                                                                            |